Article

# A high black-hole-to-host mass ratio in a lensed AGN in the early Universe

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Early JWST observations have uncovered a population of red sources that might represent a previously overlooked phase of supermassive black hole growth[1–3]. One of the most intriguing examples is an extremely red, point-like object that was found to be triply imaged by the strong lensing cluster Abell 2744 (ref. 4). Here we present deep JWST/NIRSpec observations of this object, Abell2744-QSO1. The spectroscopy confirms that the three images are of the same object, and that it is a highly reddened ($A_V \simeq 3$) broad emission line active galactic nucleus at a redshift of $z_{\rm spec} = 7.0451 \pm 0.0005$. From the width of Hβ (full width at half-maximum = 2,800 ± 250 km $s^{-1}$), we derive a black hole mass of $M_{\rm BH} = 4^{+2}_{-1} \times 10^7\ M_\odot$. We infer a very high ratio of black-hole-to-galaxy mass of at least 3%, an order of magnitude more than that seen in local galaxies[5] and possibly as high as 100%. The lack of strong metal lines in the spectrum together with the high bolometric luminosity ($L_{\rm bol} = (1.1 \pm 0.3) \times 10^{45}$ erg $s^{-1}$) indicate that we are seeing the black hole in a phase of rapid growth, accreting at 30% of the Eddington limit. The rapid growth and high black-hole-to-galaxy mass ratio of Abell2744-QSO1 suggest that it may represent the missing link between black hole seeds[6] and one of the first luminous quasars[7].

Abell2744-QSO1 (ref. 4) was discovered in JWST/Near InfraRed Camera[8] (NIRCam) pre-imaging of the Ultra-deep NIRSpec and NIRCam Observations before the Epoch of Reionization[9] (UNCOVER) program, behind the Hubble frontier fields[10] (HFFs) strong lensing cluster Abell 2744 (ref. 11) ($z = 0.308$; hereafter A2744), known to be able to significantly magnify distant background objects[12]. Abell2744-QSO1 had been identified as a UV-faint ($M_{\rm UV,1450} = -16.98 \pm 0.09$), lensed high-redshift candidate[13], but it remained an otherwise inconspicuous source in previous 0.5–1.5 μm observations with the Hubble Space Telescope (HST). The NIRCam 1–5 μm imaging showed it to be an unusually red (for example, F277W−F444W = 2.63 ± 0.10), unresolved point-source triply imaged by the foreground cluster[4], with the high-redshift nature of the source further supported geometrically by the lensing configuration. Its unique HST and JWST spectral energy distribution (SED), combined with the point-like light profile, suggested that the source may be a highly reddened active galactic nucleus (AGN) at redshift $z_{\rm phot} \approx 7$–8 (refs. 3,4).

The three images of A2744-QSO1 were recently targeted with the Near InfraRed Spectrograph[14] (NIRSpec) aboard JWST as part of the UNCOVER program (Fig. 1). Stacking the spectra of the three images of A2744-QSO1, we obtain an approximately 38 h spectrum, the deepest spectrum so far, to our knowledge, of any single high-redshift object

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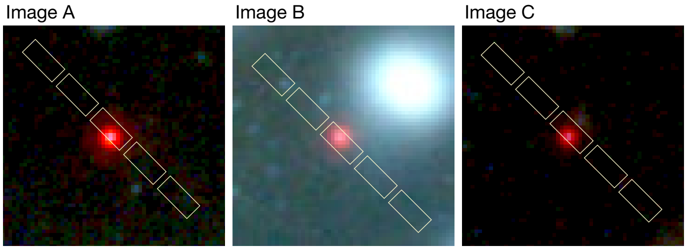


**Fig. 1 | Composite-colour image cutouts of the three images of A2744-QSO1.** The 2.4″ × 2.4″ cutouts of the UNCOVER JWST/NIRCam composite-colour image[12] (blue, F115W + F150W; green, F200W + F277W; red, F2356W + F410M + F444W) show the three images of A2744-QSO1 overlaid with the NIRSpec micro-shutter array slitlets (of 2.5″ length) of one of our micro-shutter array masks as an example (the three images were targeted with several masks, and the exact position of the object in each slit may slightly shift).

observed with JWST. After correcting for gravitational magnifications[12] of $\mu_A = 6.2^{+0.5}_{-0.2}$, $\mu_B = 7.3^{+0.1}_{-1.5}$ and $\mu_C = 3.5^{+0.2}_{-0.2}$, this corresponds to a 1.5-μm continuum depth of about 31.0 AB magnitudes at 5$\sigma$, equivalent to integrating for more than about 1,700 h in a blank field.

The spectrum shown in Fig. 2 features several strong hydrogen emission lines—a prominent Balmer series with Hα, Hβ, Hγ and Hδ, as well as Lyman-α (Lyα)—along with several faint metal lines. We give a full list of the detected emission lines in Table 1. The broad hydrogen emission lines securely confirm this object at $z_{spec} = 7.0451 \pm 0.0005$ and provide unambiguous evidence that A2744-QSO1 is an AGN. The broad-line width is measured by fitting the Hβ line, which is the clearest and most isolated because Hα is situated on the edge of the detector. We thus find an Hβ full width at half-maximum (FWHM) of 2,800 ± 250 km s$^{-1}$, well-resolved, even at the relatively low resolution of the prism (Fig. 2). These line widths are not seen in star-forming galaxies but routinely observed in the broad-line regions of AGN, in which gas clouds orbit the supermassive black hole[15]. An additional fit to the [O III] $\lambda$5008Å line shows that it is narrow and unresolved as opposed to the broad Balmer lines. Assuming a Small Magellanic Cloud extinction law[16], the Balmer line ratio (Hα to Hβ) indicates a strong dust attenuation of $A_V = 3.0 \pm 0.5$. Note that flatter attenuation curves yield higher $A_V$. The Atacama Large Millimeter/sub-millimeter Array (ALMA) 1.2 mm observations of A2744 (ref. 17) firmly rule out dusty star formation powering the Balmer lines. If caused by star formation, the emission lines would indicate an ongoing star-formation rate (SFR) of $\psi \approx 40 M_\odot$ yr$^{-1}$ and a corresponding 1.2-mm flux of about 1 millijansky (mJy), whereas no emission is observed to less than 0.1 mJy (at 3$\sigma$). We can, therefore, rule out the possibility of our source being a star-forming galaxy. However, we cannot entirely rule out that the observed rest-frame UV emission is at least partly of stellar origin. In that case, assuming the entire UV emission

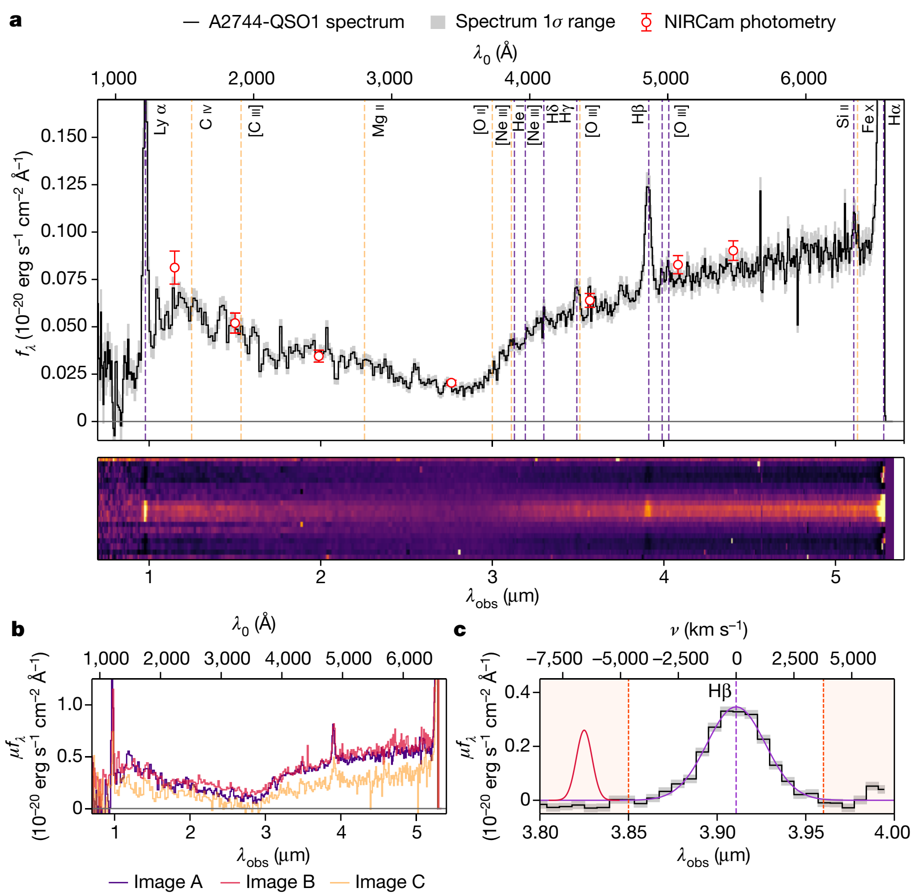


**Fig. 2 | NIRSpec-prism spectrum of A2744-QSO1. a**, The de-magnified stacked spectrum (black), equivalent to about 1,700 h integration time on target once the lensing is factored in, and its 1$\sigma$ uncertainty range (grey). Apart from its red rest-frame optical continuum[4], the source shows strong hydrogen emission lines in the Balmer series and Lyα. The emission lines enable us to precisely measure the spectroscopic redshift at $z_{spec} = 7.0451 \pm 0.0005$. We overlay the NIRCam photometry measured previously[4] as red circles. Securely identified and detected (≥3$\sigma$) emission lines are marked in purple and selected non-detected lines are shown in orange. **b**, The individual (magnified) spectra of the three images. All three images perfectly align in wavelength space, thus confirming the triply imaged nature of our object. **c**, The continuum-subtracted Hβ line (black) and our Gaussian fit to the line (purple). The red shaded areas delimit the region in which the fit is performed and the dark red curve illustrates the NIRSpec-prism LSF at the wavelength of Hβ. All errors shown represent 1$\sigma$ uncertainties.

**Table 1 | Emission lines detected in the combined spectrum of A2744-QSO1**

| Emission line | $\lambda_0$ (Å) | $\lambda_{obs}$ (μm) | $\mu F_{line}$ ($10^{-20}$ erg s$^{-1}$ cm$^{-2}$) | EW$_0$ (Å) |
|---|---|---|---|---|
| Lyα | 1,216 | 0.978 | 236.8±18.8 | 65.5±12.1 |
| Mg II | 2,796, 2,803 | 2.252 | <11 | – |
| He I | 3,889 | 3.129 | 18.3±3.3 | – |
| [Ne III] | 3,968 | 3.192 | 15.2±2.5 | – |
| Hδ | 4,102 | 3.300 | 19.0±2.4 | – |
| Hγ[a] | 4,341 | 3.492 | 34.5±3.4 | – |
| Hβ | 4,863 | 3.912 | 100.8±3.3 | 42.5±2.1 |
| [O III] | 4,960 | 3.990 | 5.0±2.6 | – |
| [O III] | 5,008 | 4.029 | 15.5±2.6 | – |
| Si II[b] | 6,347 | 5.106 | 21.3±3.2 | – |
| [N II] | 6,550 | 5.270 | 76.9±19.1 | – |
| Hα | 6,565 | 5.282 | 753.2±29.1 | 158.0±11 |

The stacked spectrum is presented in Fig. 2. Columns represent, from left to right, emission line designation, rest-frame wavelength, observed wavelength, measured line flux (not corrected for magnification) and rest-frame equivalent width (if measured). Upper limits are at 3σ. For Lyα, the continuum is based on the extrapolated rest-frame UV continuum assuming a power law.

[a]Blended with [O III]λ4363Å.

[b]Blended with [O I]λ6364Å and [Fe X]λ6375Å.

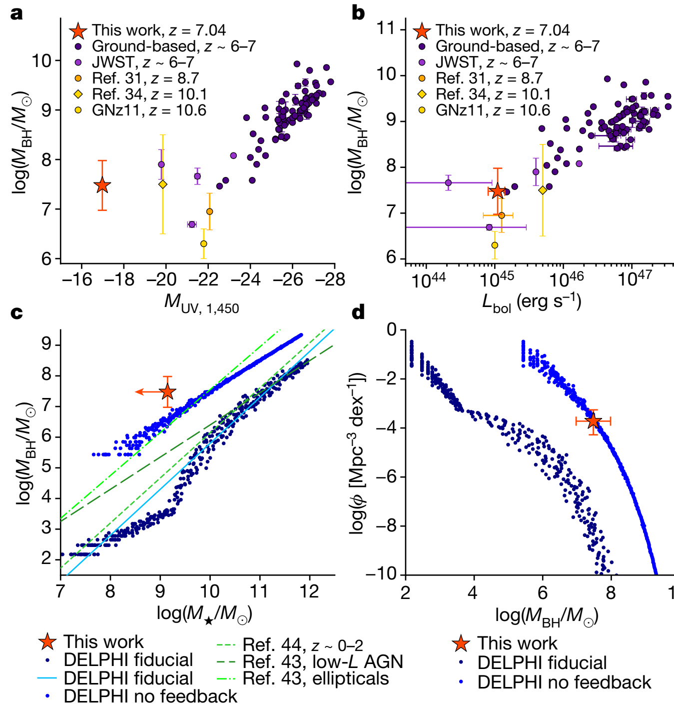


**Fig. 3 | A2744-QSO1 compared with other broad-line AGN. a,b**, A2744-QSO1 (red star) in the black hole mass–luminosity space of high-redshift broad-line AGN. The purple circles show UV-bright AGN observed from the ground[22,42], the violet circles show AGN detected with JWST at $z \sim 6$–7 (refs. 24,29,33), and the orange and yellow symbols show recently confirmed AGN at higher redshifts[31,32,34]. Although A2744-QSO1 is notably fainter in the UV than other AGN recently detected with JWST (**a**) because of its strong dust attenuation, it is typical in the $M_{BH}$–$L_{bol}$ plane (**b**). **c**,**d**, A2744-QSO1 in $M_{BH}$–$M_\star$ space (**c**) and in number densities (**d**) compared with the DELPHI simulations[37,38] (blue) and AGN observed at low redshift[43,44] (green). The dark blue points show the default DELPHI runs, which predict much higher host stellar masses and much lower number density for a black hole of this mass. The light blue points show the models with feedback turned off, in which case the models can approach the observations of A2744-QSO1. Error bars represent 1σ uncertainties.

comes from stars, we can estimate the stellar mass by assuming a constant star-formation history (SFH) and a formation redshift of $z = 12$ and obtain $M_\star \approx 10^{8.2} M_\odot$.

From the width of the Hβ line (Fig. 2), we derive the black hole mass using standard scaling relations[18] and obtain $M_{BH} = 4^{+2}_{-1} \times 10^7\ M_\odot$. We further derive a bolometric luminosity of $L_{bol} = (1.1 \pm 0.3) \times 10^{45}$ erg s$^{-1}$ (Fig. 3) based on the emission line luminosity. Both $M_{BH}$ and $L_{bol}$ are corrected for magnification. The black hole must accrete at a rate of $\dot{M} \approx 0.1 M_\odot$ yr$^{-1}$ to maintain this luminosity, which roughly corresponds to 30% of the Eddington limit—set by the balance between gravity and radiation pressure assuming spherical accretion ($L_{bol}/L_{Edd} \approx 0.3$).

We can also place a limit on the hidden stellar mass of the host galaxy based on the light profile. All three images are point sources, despite lensing magnifications differing by a factor of 2, which suggests that the intrinsic size of the source is smaller than the point spread function (PSF) of the highest magnification image. To derive formal upper limits on the size, we fit a Sérsic[19] model to the source in the (highest resolution) F115W band, finding the source to be unresolved with a size $r_e < 30$ pc (95% upper limit) after correcting for the lensing shear (equivalent to less than half the size of the PSF). Adopting an upper stellar density limit equal to that of the densest star clusters[20] or the densest elliptical galaxy progenitors[21] known to date, $\Sigma_\star \approx 5 \times 10^5 M_\odot$ pc$^{-2}$, we can derive an upper limit for how much stellar mass may be contained within $r_e < 30$ pc, thus obtaining $M_\star < 1.4 \times 10^9 M_\odot$. Our previous photometric fit, including ALMA 1.1 mm constraints returns an intrinsic stellar mass upper limit that is consistent with this estimate[3,4]. The implied $M_{BH}$–$M_\star$ ratio of greater than 3% is extreme compared with local values ($M_{BH}/M_\star \approx 0.1\%$) but similar to some other recently discovered high-redshift black holes[22].

The spectrum of A2744-QSO1 (Fig. 2) exhibits some distinct characteristics compared with typical AGN[23] and other moderate-luminosity $z > 4$ AGN discovered with JWST[24]. The optical continuum is very red, and the metal or high-ionization lines (for example, the C IV λλ1548, 1550Å, C III] λλ1907, 1909Å, Mg II λλ2796, 2803Å and the [O III] λλ4959, 5007Å doublets) are very weak compared with the hydrogen lines (Table 1 and Fig. 2). Furthermore, using the latest deep X-ray imaging obtained by the Chandra X-ray Observatory, we find that all three images are undetected in the X-rays. The combined 3σ upper limit on the rest-frame 40 keV X-ray luminosity of our source is $L_{X,40keV} < 3 \times 10^{43}$ erg s$^{-1}$, at least 10 times weaker than expected based on the UV/optical. This intrinsic X-ray faintness of A2744-QSO1 might suggest a powerful dusty wind, as is seen in low-mass rapidly accreting black holes at low redshift[25]. These dusty wind-dominated sources are rare at cosmic noon[26], however, and we do not detect any direct evidence of a strong wind in the current data. By contrast, many black holes radiating at or above their Eddington limits exhibit these unique properties[27], perhaps because of changes in the structure of their accretion disk[28]. Our estimated Eddington ratio of 0.3 would also be consistent with these characteristics.

A2744-QSO1 represents one of the first examples of a heavily reddened broad-line AGN (red dots) at high redshift[4], although hints existed in previous work also[29]. These objects are too UV-faint to have been discovered by past ground-based photometric searches[30], but many similar compact red objects have been detected in JWST surveys since[2,3,24], as well as other high-redshift AGN[1,24,31–33], including behind A2744 (ref. 34). In fact, the UNCOVER red dot sample[3] contains a second object at $z_{spec} = 7.04$ (ref. 35), consistent with A2744-QSO1, suggesting that these objects may be clustered. Three out of seven similar objects found in a blank field study[2] were also at similar redshifts to each other. Nevertheless, A2744-QSO1 remains a unique case. As the highest-redshift quasar-like object multiply imaged by a galaxy cluster known to date, it affords a remarkable signal-to-noise ratio for such a low-luminosity object.

We derive the spatial number density of A2744-QSO1-type objects by taking the sample of 'red dots' detected in UNCOVER[3,35] and verifying how many of them are confirmed at the same redshift and UV luminosity with the UNCOVER spectroscopy and updated strong lensing model. This yields a number density of $\phi = (7.3 \pm 1.7) \times 10^{-5}$ Mpc$^{-3}$ mag$^{-1}$, which is consistent with what we found previously based on the UNCOVER photometry[3] ($\phi = 2.5^{+2.6}_{-2.1} \times 10^{-4}$ Mpc$^{-3}$ mag$^{-1}$) and also with what has been found for reddened $z > 5$ AGN in JWST blank fields[2]. Although this represents only a few per cent of galaxies at this epoch[36], it is around 100 times more numerous than the faintest UV-selected AGN[30]. Because the UV luminosity of A2744-QSO1 is only $M_{\rm UV,1450} = -16.98 \pm 0.09$, finding this extremely red source at high redshifts required the unprecedented infrared sensitivity of JWST combined with gravitational lensing. As seen in Fig. 3a,b, although A2744-QSO1 is very faint in rest-frame UV compared with other high-redshift AGNs, it is nevertheless typical in terms of bolometric luminosity.

A2744-QSO1 is a significant outlier, both in terms of its $M_{\rm BH}$–$M_\star$ ratio (which is a lower limit as we do not detect the host galaxy) and the expected number density when compared with theoretical models such as the semi-analytic Dark Matter and the Emergence of Galaxies in the Epoch of Reionization (DELPHI) model[37–39]. Assuming black hole seeds with a seed mass of $150M_\odot$ originating from metal-free stars, the DELPHI model predicts that black holes as massive as A2744-QSO1 should be hosted by galaxies with stellar masses of about $M_\star \approx 10^{10.5}M_\odot$, that is, much lower $M_{\rm BH}$–$M_\star$ ratios (Fig. 3), and it underpredicts the number density of A2744-QSO1-like objects. Hydrodynamic simulations make similar predictions at $z \approx 6$ (ref. 39), although typically starting with heavier seeds ($10^4$–$10^6M_\odot$). In general, the models fail to predict the high number density associated with A2744-QSO1, which can be reproduced only by the most extreme models without feedback (Fig. 3). This conflicts with the hints of strong outflows in our source discussed above.

The discovery of A2744-QSO1 shows that our understanding of black hole growth in the early Universe is still incomplete. The photometric samples suggest high number densities, such that a few per cent of galaxies at $z > 7$ harbour reddened and rapidly growing black holes. Deep spectroscopic follow-up with JWST will be needed to confirm these number densities and explain the properties of these objects. The hints of extreme $M_{\rm BH}$–$M_\star$ ratios, of a few per cent or more, will put strong new constraints on when black holes form relative to their host galaxies. Black holes that outgrow their galaxies put strain on AGN evolution models and might hint at significant dark matter accretion onto primordial seeds[40] or at hyper-compact star-bursting clusters[41] in the early Universe. There are many outstanding puzzles about these objects remaining that JWST is only beginning to address. For A2744-QSO1 in particular, higher-resolution spectroscopy in the rest-frame UV could reveal the possible presence of a dusty outflow in this source, whereas observations at longer wavelengths could help to better constrain the mass of its host galaxy.

## Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-024-07184-8.


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## Methods

Throughout this work, we assume a standard flat ΛCDM cosmology with $H_0 = 70$ km s$^{-1}$ Mpc$^{-1}$, $\Omega_\Lambda = 0.7$, and $\Omega_m = 0.3$. Magnitudes are quoted in the AB system[45] and uncertainties represent $1\sigma$ ranges unless stated otherwise.

### JWST, ALMA and Chandra observations

The UNCOVER observations used in this work comprise both ultra-deep JWST[46,47] imaging and spectroscopy of the A2744-field with NIRCam[8] and NIRSpec[14,48] to $5\sigma$ depths of 29.9 magnitudes each[9,49]. As gravitational magnification on average adds about two magnitudes to the depth in the strong lensing regime, UNCOVER reaches effective depths of about 32 AB once the magnification is factored in. The NIRCam data cover a total area of approximately 45 arcmin$^2$ in six broadband and one medium-band filter: F115W, F150W, F200W, F277W, F356W, F410M and F444W. The data were reduced and drizzled into mosaics with the Grism redshift and line analysis software for space-based slitless spectroscopy[50] (grizli; v.1.6.0). The UNCOVER mosaics, which also include other public JWST imaging data (that is, the JWST-GLASS imaging[51] and DDT program 2756; principal investigator W. Chen; as well as archival HST imaging) and were made publicly available on the UNCOVER website. The photometry for all three images of A2744-QSO1 can be found in table 1 of the discovery paper for A2744-QSO1 (ref. 4).

The NIRSpec component of the UNCOVER survey consists of seven micro-shutter array (MSA)[52] pointings on the A2744-field with various exposure times between around 3 h and 17 h, totalling 24 h NIRSpec time and 680 targeted objects. The observations follow a two-point dither pattern and a three-shutter slitlet nod pattern at a position angle of 266°. The observations were taken from 31 July 2023 to 2 August 2023, using the low-resolution prism disperser, which covers the wavelength range of 0.7–5.3 μm at spectral resolutions $R \approx 30$–$300$. The exposure times for the three images of A2744-QSO1 ranged from 9.3 h to 16.4 h. The prism data were reduced with MSAEXP (ref. 53) (v.0.6.10). The code works on individual frames to correct for $1/f$ noise, find and mask snowballs and remove the bias. The pipeline first applies a WCS, and then identifies each slit and applies the flat-fielding and photometric corrections. The two-dimensional slits are then extracted and drizzled onto a common grid. The details of the observational setup and reduced data products, planned for public release before JWST Cycle 3, will be presented in S.H.P. et al. (manuscript in preparation). All three images of A2744-QSO1 were included in the MSA designs with cumulated exposure times of 17.4 h, 9.9 h and 14.7 h for images A, B and C, respectively.

Furthermore, A2744 is part of the ALMA Frontier Fields survey[54], which obtained the first deep 1.1 mm imaging of the cluster core. These data have now recently been complemented with additional observations covering the entire UNCOVER footprint (program ID: 2022.1.00073.S; principal investigator S. Fujimoto), achieving $1\sigma$ depths down to 32.7 μJy per beam over an area of about 24 arcmin$^2$. The UNCOVER ALMA data were also publicly released and described in detail in ref. 17.

A2744 has also been the subject of sensitive X-ray follow-up observations using the Chandra X-ray Observatory. The 1.25 Ms of Advanced CCD Imaging Spectrometer data used here were previously reduced, analysed and presented in the supplementary materials section of ref. 55. We use these data products to perform forced photometry at the locations of the three images of A2744-QSO1. Images A and B are heavily contaminated by the foreground X-ray gas associated with A2744 itself. Image C provides the cleanest (lowest background) region to perform an X-ray analysis of A2744-QSO1. However, we find no notable detection of an X-ray point source. We place a $3\sigma$ upper limit of less than 12.8 photons in the 2–10 keV energy band, that is, a flux limit of $f_{X,2-10keV} < 2.8 \times 10^{-16}$ erg s$^{-1}$ cm$^{-2}$ assuming a typical AGN X-ray power-law slope of $\Gamma = 1.9$. This is equivalent to a de-magnified X-ray luminosity of $L_{X,40keV} < 5 \times 10^{43}$ erg s$^{-1}$ at a rest-frame energy of about 40 keV given the high redshift of A2744-QSO1. Stacking and de-magnifying all three images produces a marginal increase in sensitivity, owing to the increased magnification but higher backgrounds of images A and B of $L_{X,40keV} < 3 \times 10^{43}$ erg s$^{-1}$.

### Gravitational lensing

Throughout this work, we use the most recent strong lensing model of A2744 in ref. 12, which is based on UNCOVER data and also publicly available on the UNCOVER website. The model is slightly updated here with a new multiple-image system and an additional spectroscopic redshift[56]. The model was constructed using a new implementation of the Zitrin-parametric code[12,57,58], which is not coupled to a fixed grid resolution and is thus capable of high-resolution results. For the model, the cluster galaxies are modelled each as a double pseudo-isothermal ellipsoid[59], scaled according to its luminosity using common scaling relations motivated by the fundamental plane. Three of the brightest cluster galaxies (BCGs), as well as several other galaxies situated close to multiple images, are modelled independently. We use 421 cluster galaxies throughout the UNCOVER approximately 45 arcmin$^2$ field of view, more than half of which are spectroscopically confirmed[60]. The rest were chosen based on the red-sequence of the cluster, identified in a colour–magnitude diagram based on the HST/ACS F606W and F814W bands bracketing their 4,000 Å break. Five cluster-scale dark matter halos are used, each modelled as a pseudo-isothermal elliptical mass distribution[61,62]. These dark matter halos are centred on the five BCGs but for three of them, the central position can be iterated for in the minimization process within a couple of arc seconds around the position of their respective BCG. As constraints, 141 multiple images (belonging to 48 sources) are used, 96 of which have spectroscopic redshifts. The vast majority of these are situated in the main cluster core similar to A2744-QSO1. The minimization is performed in the source plane using an approximately $10^5$-step Markov chain Monte Carlo (MCMC) process. For the minimization and uncertainty estimation, we use a positional uncertainty of 0.5″ for each multiple image. The resulting model has a very good image reproduction root mean square (RMS) of $\Delta_{RMS} = 0.51''$ in the lens plane. We infer magnifications and 95% confidence intervals of $\mu_A = 6.15$ [5.76, 6.92], $\mu_B = 7.29$ [5.11, 7.65] and $\mu_C = 3.55$ [3.31, 3.80] for images A, B and C, respectively. These are only very slightly different from the magnifications presented in ref. 4 and are in generally good agreement with the photometric flux ratios.

### Spectroscopic analysis

We use the image multiplicity of A2744-QSO1 to achieve maximum signal-to-noise ratio by stacking the weighted, local-background subtracted and co-added spectra of the three images. The spectrum is then extracted with MSAEXP using an optimal extraction[63] and collapsed to a one-dimensional spectrum. Taking the magnification into account, the effective exposure time of this stacked spectrum is about 1,700 h. The flux calibration is then performed by bootstrapping to the total photometry of image A, which has the cleanest photometric measurement[4]. Wavelength-dependent slit losses are modelled with a first-order polynomial for each MSA by convolving with the filter band pass and solving for the polynomial coefficients by comparing with the photometry. Although our stacking and flux-calibration procedures are optimal for maximizing the signal-to-noise ratio and measuring properties such as redshift and black hole mass, they unfortunately smooth out any possible variation between the three images because of the gravitational time delay. Studies of the time variability of multiply imaged AGN are important, for example, allowing for cosmography studies and reverberation mapping. This requires careful calibration of the individual spectra and goes beyond the scope of this work.

The final stacked spectrum of A2744-QSO1 displays numerous emission lines and is shown in Fig. 2. For the most robust emission line in our spectrum, the Balmer line Hβ, we use specutils[64] (v.1.10.0) to fit a two-component (broad and narrow) Gaussian profile to the continuum-subtracted emission line and estimate the uncertainties with an MCMC analysis using emcee[65] (v.3.1.4). Our line fit explicitly includes the NIRSpec point-source line-spread-function (LSF)[66] into the model at the wavelength of Hβ by convolving the model line profile with the LSF in each step of the MCMC before calculating the likelihood. The FWHM = 2,800 ± 250 km $s^{-1}$ used in the following is, therefore, intrinsic—that is, corrected for instrumental effects. Hα is more challenging to model as the red wing of the spectrum falls off the detector and is blended with [N II] $\lambda\lambda$6550, 6585Å. Moreover, the spectral resolution seems to be high enough at these wavelengths to resolve the narrow component on top of the broad wing. For completeness, we, therefore, proceed to model Hα, [N II] $\lambda\lambda$6550, Hβ and [O III] jointly, fixing the metal lines to 100 km $s^{-1}$ FWHM, restricting the narrow-to-broad line ratio to be the same for Hβ and Hα, but allowing for different line widths between Hβ and Hα. The fit parameters are well constrained and reproduce the broad-line width of Hβ, while finding a smaller FWHM for the broad component of Hα FWHM = 2,300 ± 250 km $s^{-1}$ as is commonly seen in low-redshift AGN[18]. For consistency, we estimate the width of the narrow lines by fitting the [O III] $\lambda$5008Å line with the same method as above and obtain an LSF-corrected line width that is well below the width of the LSF. This shows that the forbidden lines are narrow. Other spectral lines are identified and modelled with single Gaussians and presented in Table 1. Exceptions to this are Hγ, which is blended with [O III] $\lambda$4363Å, and Si II $\lambda$6347Å, which is blended with [O I] $\lambda$6364Å and [Fe X] $\lambda$6375Å. These are modelled similarly to Hα, that is, including the blended emission lines, the LSF and the fixed narrow line width. Note that Hγ also has a resulting broad FWHM, consistent with Hα and Hβ.

We derive a Balmer decrement of 7.4 ± 0.4 from the Balmer line measurements in Table 1. Assuming a reddening law based on fits to the Small Magellanic Cloud[16] and for $R_V = 2.72$ (refs. 67,68), we find $A_V = 3.0 \pm 0.5$, $A(\mathrm{H}\beta) = 3.1 \pm 0.5$ and $A(\mathrm{H}\alpha) = 2.1 \pm 0.5$. To date, there have been no direct measurements of the dust attenuation law in obscured high-redshift AGN. The Small Magellanic Cloud extinction law has, however, been found to match low-redshift dusty AGN best[69,70] and also matches low-metallicity high-redshift galaxies well[71–74]. We note that if we assumed a flatter reddening law, the reddening corrections would be larger as would the bolometric luminosity[75]. For example, assuming a Calzetti attenuation law[76], we obtain $A_V = 4.5$. The line fluxes are corrected for dust attenuation before black hole mass and Eddington ratio calculations are made. Although the Balmer decrement in the broad-line region arises in part within the broad-line region itself[77], we note that we derive very comparable $A_V$ from the photometric modelling performed in a previous work[3,4].

To investigate a possible SFR origin of the Balmer lines, we convert the dust-corrected Hα luminosity to an SFR (ref. 78), resulting in an ongoing SFR of $\psi \approx 40 M_\odot$ $yr^{-1}$ and predict the rest-frame far-infrared flux using the Flexible Stellar Population Synthesis code[79], which implements the dust model in ref. 80. The predicted ALMA 1.2 mm fluxes are $f_{1.2\mathrm{mm}} \approx 1$ mJy in contrast with a 3$\sigma$ upper limit of less than 0.1 mJy based on the deep ALMA map, ruling out the possibility of star formation as the origin of the emission lines.

We use the Hβ and Hα line widths and attenuation-corrected line fluxes to estimate the single-epoch black hole mass. Because we do not measure the intrinsic continuum, we use the de-reddened and de-magnified Balmer line luminosity to calculate $M_{\mathrm{BH}}$ (ref. 18). We find a black hole mass of $M_{\mathrm{BH}} = 4^{+2}_{-1} \times 10^7 M_\odot$, taking the average of Hα and Hβ, which are consistent with each other. The errors reflect both the line width uncertainties, the differences between the Hα and Hβ measurement and the dust attenuation uncertainty. The additional systematic uncertainty term on single-epoch black hole masses is estimated to be about 0.5 dex (ref. 81). We find a bolometric luminosity of $L_{\mathrm{bol}} = (1.1 \pm 0.3) \times 10^{45}$ erg $s^{-1}$, assuming a bolometric correction of 10 ± 2 (ref. 82), based on the emission line luminosities. For consistency, we also derive the bolometric luminosity from the continuum at rest-frame 5,100 Å and find a consistent result. The continuum luminosity estimated from the broadband photometry in ref. 4 is higher than this estimate by nearly a factor of 10, resulting from uncertainties in quantities such as the redshift and the reddening. We thus infer an Eddington ratio of $L_{\mathrm{bol}}/L_{\mathrm{Edd}} \approx 0.3$.

### Size measurement

In ref. 4, the authors used galfit[83] measurements to the individual images of A2744-QSO1, in all available broadbands from F115W to F444W, together with the fact that all three images appear as point sources despite suffering different amounts of shear and magnification, to conclude that the object was point-like and from the PSF size to obtain a source size estimate of $r_e < 35$ pc (ref. 4).

Here we revisit this size estimate with the most up-to-date PSF models and data reduction. Using pysersic[84], we fit a single Sérsic model to the F150W image. We find a firm upper limit on the size of $r_e < 11.2^{+1.6}_{-0.8}$ mas, which translates to a 2$\sigma$ upper limit of $r_e \lesssim 100$ pc for the lensed image. Applying a lensing correction of $\mu_t = 3.52$—the tangential shear in image A used for the size measurement—we then confirm a firm 2$\sigma$ upper limit of 30 pc on the source radius, in agreement with the measurement in ref. 4.

### Number density

To derive the number density of objects such as A2744-QSO1, we need to compute the corresponding effective co-moving volume $V_{\mathrm{eff}}$, which folds in the lensing distortion and possible selection effects. This is done following a forward-modelling approach[36], which was re-implemented for JWST observations and in particular for UNCOVER[85]: We use the (de-magnified) SED of A2744-QSO1 to populate the UNCOVER source plane of A2744[12] with 10,000 similar mock sources. These are then deflected into the lens plane and injected in the UNCOVER mosaics on which we re-run the source detection and selection methods used for the catalogs[49,86]. The ratio of recovered to injected sources enables us to determine the selection function for objects such as A2744-QSO1. This selection function is multiplied with the co-moving volume element that is then integrated over the source-plane area of sufficient magnification that enables us to detect the source at 3$\sigma$ in each NIRCam band, given the UNCOVER mosaic depths[49]. The effective volume thus obtained for A2744-QSO1 is $V_{\mathrm{eff}} = 27541 \pm 294$ $\mathrm{Mpc}^3$.

Given that the field of view of about 45 $\mathrm{arcmin}^2$ of the UNCOVER survey[9] contains two 'red dot' objects of the redshift and UV luminosity of A2744-QSO1[3], A2744-QSO1 itself and one other[35] projected 369 kpc away in the source plane, the number density is $\phi = (7.3 \pm 1.7) \times 10^{-5}$ M $\mathrm{pc}^{-3}$ $\mathrm{mag}^{-1}$.

### Theoretical expectation

We compare our results with the Dark Matter and the Emergence of Galaxies in the Epoch of Reionization[37,38] (DELPHI) semi-analytic model. This is suited to jointly track the assembly of the dark matter, baryonic and black hole components of high-redshift ($z \gtrsim 7$) galaxies up to $z = 40$. Starting at $z \sim 7$, we build analytic (binary) merger trees for 700 galaxies equally spaced in log-space in the halo mass range $M_h = 10^8$–$10^{15} M_\odot$ using a constant time step of 30 Myr. Each of these halos is assigned a number density according to the Sheth–Tormen halo mass function[87] at $z \sim 7$ and the number density propagated throughout the merger tree.

In this work, we consider only seeding by stellar black hole seeds formed from metal-free (Population III) stars. The first halos (starting leaves) of any halo are seeded with a stellar black hole (of $150 M_\odot$) at $z > 13$ assuming that halos collapsing from high ($\gtrsim$3.5)$\sigma$ fluctuations in the primordial density field are most likely to host these seeds. At any time step, the initial gas mass is the sum of that brought in by mergers and smooth accretion. A fraction of this gas can form stars with an efficiency that is the minimum between the supernovae (SN II) energy required to unbind the rest of the gas and an upper threshold $f_*$. In terms of black holes, in our fiducial model, these grow both through (instantaneous) mergers and accretion. A fraction of the gas mass left in the halo after star formation and its associated feedback can be accreted onto the black hole with the accretion rate depending on a critical halo mass (which at $z \sim 7$ corresponds to $10^{10.9} M_{\odot}$). Although black holes in lower mass halos are assumed to have puffed-up disks, allowing for only low accretion rates (about $7.5 \times 10^{-5}$ Eddington), black holes in halos above this critical mass can accrete a fixed fraction of the gas mass up to the Eddington limit; a fraction of the black hole energy (0.3%) is allowed to couple to the gas content. The maximal (no feedback) model presents an upper limit in which every black hole is always allowed to accrete the gas mass available, up to the Eddington limit, and we ignore gas mass lost in outflows, both due to type II supernova and black hole feedback.

The model has been tuned to reproduce all the key observables for both star-forming galaxies at $z \sim 5$–10 (including the evolving UV luminosity and stellar mass functions) and AGN at $z \sim 5$–6 (including the AGN UV luminosity and black hole mass functions).

## Data availability

The raw JWST data used in this work are publicly available on the Barbara A. Mikulski Archive for Space Telescopes (MAST). These observations specifically are associated with the JWST GO program no. 2561, JWST ERS program no. 1324, and JWST DD program no. 2756. The reduced UNCOVER mosaics, catalogues, lens models as well as spectra are part of the public data release by the UNCOVER team[9,12,49] (S.H.P. et al., manuscript in preparation). The public ALMA data can be found on the ESO Science Archive under IDs 2018.1.00035.L and 2013.1.00999.S. The deep Chandra image used here comprises data from more than 60 individual programs that are available on the Chandra Data Archive. The individual observations IDs are listed in table 1 of ref. 55. Source data are provided with this paper.

## Code availability

This research made use of Astropy, a community-developed core Python package for Astronomy[88,89] as well as the packages NumPy[90], SciPy[91], Matplotlib[92] and the MAAT Astronomy and Astrophysics tools for MATLAB[93].


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